SET 1 – VIVA VOCE

1. What is the difference between cell delay and net delay in static timing analysis?
2. Why is the set_input_delay constraint required during ASIC timing analysis?
3. What is the role of the always @(posedge clk) construct in sequential RTL design?
4. How does clock gating help reduce power consumption in an ASIC?
5. What is the significance of cell utilization during floorplanning?

SET 2 – VIVA VOCE

1. What is the difference between maximum delay and minimum delay analysis in STA?
2. Why is the set_output_delay constraint specified in an SDC file?
3. How does constant propagation help optimize a synthesized circuit?
4. What is the purpose of a reset signal in an FSM-based controller?
5. Why is placement optimization performed before clock tree synthesis?

SET 3 – VIVA VOCE

1. What is the difference between clock latency and clock transition time?
2. Why is the set_max_transition constraint important during synthesis?
3. How does common subexpression elimination reduce hardware resources?
4. What is the purpose of a delay register in an FIR filter?
5. Why is IR drop analysis important in ASIC physical design?

SET 4 – VIVA VOCE

1. What is the significance of the critical path in determining circuit performance?
2. Why is the set_max_fanout constraint used during logic synthesis?
3. How does retiming improve the performance of a sequential circuit?
4. What is the difference between Moore and Mealy FSM architectures?
5. What is the purpose of filler cells in the physical design flow?

SET 5 – VIVA VOCE

1. What is the difference between data arrival time and data required time in STA?
2. Why is the set_load constraint applied to output ports during synthesis?
3. How does logic duplication help improve timing performance?
4. What is the difference between blocking and non-blocking assignments in Verilog?
5. Why is electromigration analysis necessary for reliable ASIC operation?

SET 6 – VIVA VOCE

1. What is the significance of Worst Negative Slack (WNS) in a timing report?
2. Why is the set_driving_cell constraint used during synthesis?
3. How does operand isolation reduce unnecessary switching activity?
4. What is the difference between synchronous and asynchronous reset in sequential circuits?
5. What is the purpose of tie-high and tie-low cells in ASIC implementation?

SET 7 – VIVA VOCE

1. What is the significance of Total Negative Slack (TNS) in timing closure?
2. Why are setup and hold timing checks performed using different process corners?
3. How does dead-code elimination improve the synthesized circuit?
4. What is the purpose of the DONE state in an FSM-based arithmetic controller?
5. Why are decoupling capacitors inserted in ASIC power distribution networks?

SET 8 – VIVA VOCE

1. What is the difference between launch clock latency and capture clock latency?
2. Why is the set_clock_groups -asynchronous constraint used in multi-clock designs?
3. How does changing a standard cell from X1 to X4 affect its driving capability?
4. What is the significance of a comparison point in Cadence Conformal LEC?
5. What is the purpose of routing congestion analysis during physical design?

SET 9 – VIVA VOCE

1. What is the difference between cell setup time and cell hold time in a timing library?
2. Why is a set_multicycle_path -hold constraint required when defining multicycle timing paths?
3. How does FSM state encoding influence the area and timing of synthesized hardware?
4. What is the purpose of the read_libs command in Cadence Genus?
5. Why are metal density rules checked during physical verification?

SET 10 – VIVA VOCE

1. What is the significance of clock slew in sequential circuit timing?
2. Why is the set_false_path constraint applied only to functionally justified paths?
3. How does the number of pipeline stages influence the latency of a datapath?
4. What is the difference between functional equivalence and structural equivalence in ASIC verification?
5. Why is timing sign-off performed after parasitic extraction?